Unified Classification Framework for Multipolarization and Dual-Frequency SAR

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***Abstract*— For synthetic aperture radar (SAR), multipolarization and multifrequency modes greatly enrich the acquired earth resource information and have been widely applied in remote-sensing fields. In this article, we compare the classification capabilities of multipolarization and dual-frequency SAR. To meet the objective of selecting consistent and complete polarimetric information, a unified classification framework is proposed. In the framework, covariance matrices are used directly as inputs instead of polarimetric indicators. Additionally, the Wishart mixture model (WMM) is utilized to characterize the statistical distribution of polarimetric SAR data. Besides, the data log-likelihood function is utilized to mitigate the influence of the initial values on the expectation-maximization (EM) algorithm. Then, among the combinations of four sample-to-subclass distances and two schemes for obtaining sample-to-class distances, the one with the best classification performance is selected as the default for this framework. In the experiments, the classification capabilities of full polarization (FP), compact polarization (CP), and dual-polarization (DP) modes are first compared through the proposed classification framework. Then, we compare the classification capabilities of dual-frequency SAR in FP, CP, and DP modes. For polarimetric SAR (PolSAR) system design, it is necessary to strike a balance between demand indexes (classification performance, coverage width, and so on) and cost (such as budget, weight, and so on). The comparison results provide a reference for the optimization of polarization modes and frequency bands of the existing multipolarization and dual-frequency SAR payloads and the design of future SAR systems.**

***Index Terms*— Classification capability, comparison, dual-frequency, multipolarization, synthetic aperture radar (SAR), system design, Wishart mixture model (WMM).**

I. Introduction

COMPARED with optical payloads, synthetic aperture radar (SAR) has all-day, all-weather, and active imaging capability. Polarimetric SAR (PolSAR) not only inherits the advantages of SAR, but also can provide scattering information in multiple polarization directions. Therefore, it is widely applied in remote-sensing fields, such as land-use classification [1], [2], wetland mapping [3], and disaster monitoring [4]. In recent years, PolSAR systems have undergone an evolution from single polarization (SP) to dual polarization (DP) and finally to full polarization (FP). With the increase of polarimetric channels, the information obtained by the PolSAR system becomes more and more abundant. The FP system is considered to be able to obtain complete polarization information of ground targets. However, the FP system has constraints compared with the DP and SP systems, such as high transmit average power and complicated system design [5]. To balance some of the constraints of FP, compact polarization (CP) was developed. Currently, there are three working modes for the CP system [6]: $\pi/4$ [7], dual circular polarization (DCP) [8], and hybrid CP (HCP) [9]. Besides, many spaceborne and airborne PolSAR systems in different frequency bands have been successfully launched, such as C-band GF-3 [10], L-band ALOS-2 [11], L- and C-band EMISAR [12], and P-, L-, and C-band AIRSAR [13]. In the future, L- and S-band NISAR will also be launched to study ice deformation, ecosystems, and solid earth [14]. Multifrequency PolSAR systems obtain scattering information of ground objects at different wavelengths and different penetration depths. So, multifrequency PolSAR improves upon the performance of single-frequency PolSAR.

The SAR system design is affected by the balance between information quality requirements (more frequencies/polarization channels) and cost (resource use requirements). The multifrequency FP SAR system is undoubtedly the optimal choice in terms of performance. However, constraint factors such as payload size and weight, budget, transmit average power, and complicated system design often hinder the realization of the multifrequency FP SAR system, especially for space-borne systems. For specific applications, the optimization of polarization modes and frequency bands can give full play to the advantages of different polarization systems and frequency bands if the multifrequency FP SAR system is not possible. In addition, if the performance improvement of the multifrequency FP mode is not noticeable under the premise of sacrificing system indicators such as coverage width and resolution, it is not worthwhile to choose this mode. Therefore, it is necessary to compare the application potential of multipolarization and multifrequency SAR.

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Classification is an important research topic in SAR applications. For comparing the classification capabilities of multipolarization and multifrequency SAR, the commonly used method can be roughly summarized in two steps [15], [16], [17], [18], [19]. The first step is to extract polarimetric indicators from the covariance/coherency matrix. The polarimetric indicators are extracted from each frequency band for multifrequency SAR data. Then, the extracted polarimetric indicators are used as input for machine-learning or deep-learning methods to obtain classification results. The success of these methods can be further consolidated by avoiding some of the concerns about rigor and consistency. One concern is that the extracted polarimetric indicators do not represent all the information contained in the PolSAR data. For example, one polarimetric indicator was extracted from FP, CP, and DP SAR data, respectively, for classification in [15]. Obviously, the information obtained by the three PolSAR systems cannot be fully characterized by one indicator. In [19], only the backscattering coefficients were used to classify multifrequency FP SAR data. Although the experimental result showed that the classification performance of multifrequency SAR is better than single-frequency SAR, the polarization information of different frequency bands was not fully utilized. Another concern is that the polarimetric indicators utilized for classification often vary. In some articles [16], [17], [18], [20], a large number of polarimetric indicators were utilized to ensure that these indicators can adequately represent the information obtained by the PolSAR system. However, the selection of polarimetric indicators can always be more objective and consistent. For example, 25, 21, and 13 indicators were utilized for the FP, CP, and DP modes, respectively, in [16]. However, 19 FP, 15 CP, and 6 DP indicators are utilized in [20]. A large number of commonly used polarimetric indicators may tend to be highly correlated. Even with the same classification method, the change in the number of input indicators can affect the classification results [17], [18]. Besides, the classification performance does not always improve with more polarimetric indicators used, and sometimes even decreases [21]. This means that the comparison results of classification capabilities may be affected by the number of input indicators. Generally, there are some debatable factors for the classification method based on polarimetric indicators.

PolSAR systems directly record the complete polarimetric information through the Sinclair matrix and the corresponding covariance/coherence matrix. Therefore, these scattering matrices can be used to compare the classification capabilities [19], [22], [23]. Lee et al. [22] assessed the classification capabilities of multifrequency and multipolarization SAR data based on the assumption that the covariance/coherence matrix follows the Wishart distribution. Ainsworth et al. [23] adopted the Wishart maximum likelihood classifier to compare the classification capabilities of FP, CP, and DP modes. However, the Wishart model works best to describe PolSAR data in homogeneous regions, since this model does not include any texture information. For PolSAR data in complicated scenes, the Wishart model may not characterize its distribution well. There may be a considerable error between the real distribution and the statistical model, resulting in inaccurate quantitative comparison results. Based on this, more effective statistical models are needed to describe the distribution of PolSAR data. Besides, PolSAR data in different frequency bands are assumed statistical independence when evaluating the classification capabilities of multifrequency PolSAR in [22], [24], and [25]. This does not take advantage of the cross-correlation information between different frequency bands, limiting the classification capability of multifrequency PolSAR data.

In this article, we propose a unified classification framework for multipolarization and dual-frequency SAR. The framework is directly based on the covariance matrix for classification tasks without using polarimetric indicators. Thus, the consistency of the classification method is improved. Then, the Wishart mixture model (WMM) is utilized to characterize the statistical distribution of PolSAR data efficiently and flexibly. This improves agreement between the statistical model, PolSAR images, and ground-reference data. To deliver an impartial and fair assessment of the classification capabilities of multipolarization and dual-frequency PolSAR data, the cross-correlation information between different frequency bands should be retained, as is done in this framework. It should be emphasized that the proposed unified framework does not focus on the improvement of classification performance, but on the consistency and completeness of the classification method for multipolarization and dual-frequency PolSAR.

The rest of this article is organized as follows. Section II first gives the basic form of PolSAR data and then introduces the statistical model and distance measures. Section III presents the classification process of the proposed unified framework. Classification results and specific analysis are given in Section IV, and Section V concludes this article.

II. Preliminaries

A. PolSAR Data Description

FP SAR system can obtain complete scattering information of ground targets through four linear polarimetric combinations. The information is usually described in the form of a 2×2 Sinclair matrix $\mathbf{S}$, which is

$$\mathbf{S} = \begin{bmatrix} S_{\mathrm{HH}} & S_{\mathrm{HV}} \\ S_{\mathrm{VH}} & S_{\mathrm{VV}} \end{bmatrix} \tag{1}$$

where H is the horizontal polarization and V is the vertical polarization. Under the reciprocity condition, that is, $S_{\mathrm{HV}} = S_{\mathrm{VH}}$, the Sinclair matrix can be transformed into the lexicographic scattering vector, as follows:

$$\mathbf{k} = \begin{bmatrix} S_{\mathrm{HH}} & \sqrt{2}S_{\mathrm{HV}} & S_{\mathrm{VV}} \end{bmatrix}^T \tag{2}$$

where the superscript T represents the matrix transpose. To mitigate the effect of speckle, scattering vectors are usually taken in multilook processing. The covariance matrix is obtained after multilook processing on the lexicographic scattering vector, according to the following:

$$\mathbf{Z} = \frac{1}{L}\sum_{i=1}^{L} \mathbf{k} \cdot \mathbf{k}^H \tag{3}$$

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where L is the number of looks and the superscript H represents the conjugate transpose. For CP and DP SAR systems, the corresponding data can be extracted from FP data. Take the HCP SAR system as an example, which transmits circularly polarized radar waves and simultaneously receives echo in H and V polarization directions. The received echo is often written as the Jones vector, which is

$$\begin{aligned}\mathbf{k}_{\text{hcp}} &= \begin{bmatrix} S_{\text{HC}} \\ S_{\text{VC}} \end{bmatrix} = \begin{bmatrix} S_{\text{HH}} & S_{\text{HV}} \\ S_{\text{VH}} & S_{\text{VV}} \end{bmatrix} \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ \pm j \end{bmatrix} \\ &= \frac{1}{\sqrt{2}} \begin{bmatrix} S_{\text{HH}} \pm j S_{\text{HV}} \\ S_{\text{VH}} \pm j S_{\text{VV}} \end{bmatrix} \end{aligned} \tag{4}$$

where C represents the circular polarization, the "+" sign indicates left-hand circular polarization, and the "−" sign indicates right-hand circular polarization. The relationship between $\mathbf{k}$ and $\mathbf{k}_{\text{hcp}}$ is as follows:

$$\begin{aligned}\mathbf{k}_{\text{hcp}} &= \frac{1}{2} \begin{bmatrix} \sqrt{2} & \pm j & 0 \\ 0 & 1 & \pm\sqrt{2} j \end{bmatrix} \cdot \mathbf{k} \\ &= \mathbf{A} \cdot \mathbf{k} \end{aligned} \tag{5}$$

where $\mathbf{A}$ is the coefficient matrix establishing the relationship between $\mathbf{k}$ and $\mathbf{k}_{\text{hcp}}$. The covariance matrix of HCP can be obtained by

$$\mathbf{Z}_{\text{hcp}} = \frac{1}{L} \sum_{i=1}^{L} \mathbf{k}_{\text{hcp}} \cdot \mathbf{k}_{\text{hcp}}^{H} = \mathbf{A}\mathbf{Z}\mathbf{A}^{H}. \tag{6}$$

The covariance matrix of other CP and DP modes can be derived by modifying the coefficient matrix $\mathbf{A}$ in (6). For dual-frequency FP PolSAR data, Ferro-Famil et al. [26] proposed the multidimensional scattering vector $\mathbf{w}$

$$\mathbf{w} = \begin{bmatrix} \mathbf{k}_1 \\ \mathbf{k}_2 \end{bmatrix} \tag{7}$$

where $\mathbf{k}_1, \mathbf{k}_2$ are the FP scattering vectors of two frequency bands. After multilook processing, the corresponding covariance matrix $\mathbf{C}$ is

$$\mathbf{C} = \frac{1}{L} \sum_{i=1}^{L} \mathbf{w} \cdot \mathbf{w}^{H} = \begin{bmatrix} \mathbf{C}_{11} & \mathbf{C}_{12} \\ \mathbf{C}_{21} & \mathbf{C}_{22} \end{bmatrix} \tag{8}$$

where $\mathbf{C}_{mn} = (1/L) \sum_{i=1}^{L} \mathbf{k}_m \cdot \mathbf{k}_n^{H}$ $m, n = 1, 2$. The matrixes $\mathbf{C}_{11}$ and $\mathbf{C}_{22}$ are the covariance matrices for single-frequency data. The matrix $\mathbf{C}_{12} = \mathbf{C}_{21}^{\text{H}}$ contains the cross-correlation information between the two frequency bands. The relationship between $\mathbf{w}$ and the dual-frequency HCP scattering vector $\mathbf{w}_{\text{hcp}}$ is as follows:

$$\begin{aligned}\mathbf{w}_{\text{hcp}} &= \begin{bmatrix} \mathbf{k}_{\text{hcp1}} \\ \mathbf{k}_{\text{hcp2}} \end{bmatrix} = \begin{bmatrix} \mathbf{A} & 0 \\ 0 & \mathbf{A} \end{bmatrix} \cdot \begin{bmatrix} \mathbf{k}_1 \\ \mathbf{k}_2 \end{bmatrix} \\ &= \mathbf{B} \cdot \mathbf{w}. \end{aligned} \tag{9}$$

According to the coefficient matrix $\mathbf{B}$ in (9), the covariance matrix of dual-frequency HCP can be obtained by

$$\mathbf{C}_{\text{hcp}} = \frac{1}{L} \sum_{i=1}^{L} \mathbf{w}_{\text{hcp}} \cdot \mathbf{w}_{\text{hcp}}^{H} = \mathbf{B}\mathbf{C}\mathbf{B}^{H}. \tag{10}$$

B. Statistical Model and Distance Measures

For homogeneous regions, the lexicographic scattering vector is usually considered to satisfy the multivariate complex Gaussian distribution with zero mean [27], [28]. Then, the corresponding covariance matrix $\mathbf{Z}$ obeys the complex Wishart distribution, and its probability density function (PDF) is [29]

$$p(\mathbf{Z}|\mathbf{\Sigma}) = \frac{L^{Ld}|\mathbf{Z}|^{L-d}}{\Gamma_d(L)|\mathbf{\Sigma}|^{L}} \exp\left\{-L \cdot \text{tr}\left(\mathbf{\Sigma}^{-1}\mathbf{Z}\right)\right\} \tag{11}$$

where $\mathbf{\Sigma}$ represents the cluster center of $\mathbf{Z}$, d is the matrix dimension, and the operators $|\cdot|$ and $\text{tr}(\cdot)$ are the determinant and trace of the matrix, respectively. $\Gamma_d(L)$ in (11) is defined as

$$\Gamma_d(L) = \pi^{d(d-1)/2} \prod_{j=1}^{d} \Gamma(L - j + 1) \tag{12}$$

where $\Gamma(\cdot)$ is the gamma function. According to the properties of multivariate complex Gaussian distribution and (5), we know that the vector $\mathbf{k}_{\text{hcp}}$ also satisfies the multivariate complex Gaussian distribution with zero mean. Therefore, the corresponding covariance matrix $\mathbf{Z}_{\text{hcp}}$ obeys the complex Wishart distribution, and its PDF is as follows:

$$p\left(\mathbf{Z}_{\text{hcp}}|\mathbf{\Lambda}\right) = \frac{L^{Ld}|\mathbf{Z}_{\text{hcp}}|^{L-d}}{\Gamma_d(L)|\mathbf{\Lambda}|^{L}} \exp\left\{-L \cdot \text{tr}\left(\mathbf{\Lambda}^{-1}\mathbf{Z}_{\text{hcp}}\right)\right\} \tag{13}$$

where $\mathbf{\Lambda}$ represents the cluster center of $\mathbf{Z}_{\text{hcp}}$. For other CP and DP modes, the covariance matrix also obeys the complex Wishart distribution.

A distance measure represents degrees of difference between data. For PolSAR classification, a distance measure can evaluate the similarity or dissimilarity between a sample and a cluster center [30], [31], [32], [33], [34]. Therefore, distance measures can be used for PolSAR classification. A well-defined distance measure should satisfy three properties: nonnegativity, definiteness, and symmetry [32]. Next, four distance measures will be introduced, all of which are considered in the proposed unified framework. Although these distance measures are derived from FP SAR data, they also apply to CP and DP SAR data.

One distance measure is derived by taking the logarithm of the likelihood function. Lee et al. [29] obtained the Wishart distance from the log-likelihood function of the Wishart distribution. The Wishart distance is a widely used distance measure and its expression is as follows:

$$d_W\left(\mathbf{\Sigma}_x, \mathbf{\Sigma}_y\right) = \ln|\mathbf{\Sigma}_y| + \text{tr}\left(\mathbf{\Sigma}_y^{-1}\mathbf{\Sigma}_x\right) \tag{14}$$

where $\mathbf{\Sigma}_x, \mathbf{\Sigma}_y$ are two covariance matrices. When $\mathbf{\Sigma}_x, \mathbf{\Sigma}_y$ is the cluster centers of classes x and y, d_W represents the distance from class x to class y. When $\mathbf{\Sigma}_x$ is the covariance matrix of a sample and $\mathbf{\Sigma}_y$ is the cluster centers of a class, d_W represents the distance from the sample to the class. The Wishart distance is a simple and effective distance measure, but it does not satisfy the three properties mentioned above [32].

Kersten et al. [33] define a distance measure for covariance matrices based on hypothesis testing. The null versus alternative hypothesis test is: $H_0 : \mathbf{\Sigma}_x = \mathbf{\Sigma}_y$ versus $H_1 : \mathbf{\Sigma}_x \neq \mathbf{\Sigma}_y$.

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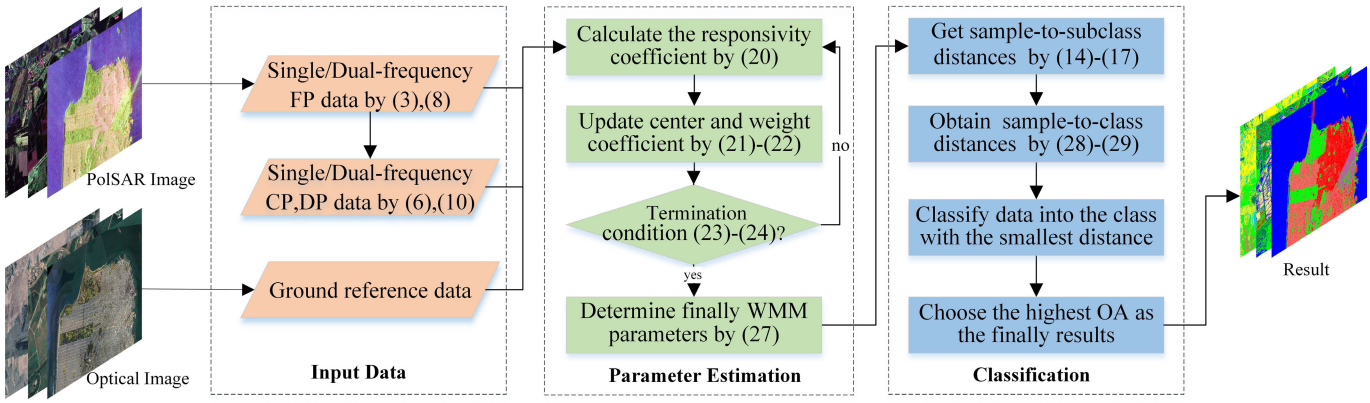


Fig. 1. Unified classification framework.

Based on the maximum ratio test and the Wishart distribution, the derived distance measure is called the Bartlett distance

$$d_{\mathrm{B}}(\boldsymbol{\Sigma}_x, \boldsymbol{\Sigma}_y) = 2\ln\frac{|(\boldsymbol{\Sigma}_x + \boldsymbol{\Sigma}_y)/2|}{\sqrt{|\boldsymbol{\Sigma}_x||\boldsymbol{\Sigma}_y|}} = 2\ln\frac{|\boldsymbol{\Sigma}_x + \boldsymbol{\Sigma}_y|}{\sqrt{|\boldsymbol{\Sigma}_x \cdot \boldsymbol{\Sigma}_y|}} - 2d\ln 2. \tag{15}$$

Furthermore, when it is assumed that $\boldsymbol{\Sigma}_y$ is known, the derived distance measure is called the revised Wishart (RW) distance

$$d_{\mathrm{RW}}(\boldsymbol{\Sigma}_x, \boldsymbol{\Sigma}_y) = \frac{|\boldsymbol{\Sigma}_x|}{|\boldsymbol{\Sigma}_y|} + \mathrm{tr}\left(\boldsymbol{\Sigma}_x^{-1}\boldsymbol{\Sigma}_y\right) - d. \tag{16}$$

The Bartlett distance satisfies the three properties mentioned above; however, the RW distance just satisfies nonnegativity, definiteness, but not symmetry. Therefore, a symmetric RW (SRW) distance is defined as [34]

$$d_{\mathrm{SRW}}(\boldsymbol{\Sigma}_x, \boldsymbol{\Sigma}_y) = \frac{1}{2}\left(\mathrm{tr}\left(\boldsymbol{\Sigma}_x^{-1}\boldsymbol{\Sigma}_y\right) + \mathrm{tr}\left(\boldsymbol{\Sigma}_y^{-1}\boldsymbol{\Sigma}_x\right)\right) - d. \tag{17}$$

d_{SRW} is also known as the Kullback–Leibler distance, which is derived from the Kullback–Leibler divergence theory. It represents the logarithmic difference between two covariance matrices that obey the Wishart distribution. It should be emphasized that the Wishart, Bartlett, RW, and SRW distance measures are all derived under the assumption that the data obey the complex Wishart distribution. Many other distance measures are available including the Bhattacharyya, Hellinger, and Rényi distances [35].

III. Unified Classification Framework

The unified classification framework is shown in Fig. 1. In the classification framework, the input data is the covariance matrix, not indicators derived from PolSAR data. In this way, some debatable factors, such as the inconsistent selection of polarimetric indicators, can be avoided. Furthermore, WMM is utilized to efficiently and flexibly characterize the statistical distribution of the input data. Compared with the complex Wishart model, WMM can greatly reduce the modeling error between the statistical model and the true distribution. Then, the expectation-maximization (EM) algorithm and the data log-likelihood function are utilized to estimate the parameters of WMM. Next, PolSAR data are classified by distance measures, which are obtained according to the parameters of WMM. Since the Wishart, Bartlett, RW, and SRW distance measures are all derived under the condition that the data obey the complex Wishart distribution, they cannot be used directly for classification. We propose two schemes to obtain distance measures for the data obeying the WMM distribution. Next, we will introduce the unified classification framework in detail.

A. WMM, EM Algorithm, and Initialization

The complex Wishart distribution is a simple and effective statistical model that has been used extensively over the past few decades. However, the complex Wishart distribution assumes that the backscattering characteristics of pixels belonging to the same class are homogeneous. This is not the case for complicated scenarios. There is considerable modeling error when the Wishart model is used to describe PolSAR data in heterogeneous regions. Therefore, more flexible and efficient statistical models are needed to mitigate the impact of the modeling error on comparison results. Gao et al. [36] proposed WMM to characterize the distribution characteristics of PolSAR data. WMM considers that there may be multiple subclasses in a single class and each subclass still follows the Wishart distribution. Each subclass has a weight coefficient which represents the prior probability that the data belongs to that subclass. From a mathematical point of view, the idea of WMM is to decompose the sophisticated distribution of real PolSAR data into several simple Wishart distributions. The PDF of WMM can be obtained by weighted summation of multiple Wishart distributions, as follows:

$$p(\mathbf{Z}|K, \pi_1, \ldots, \pi_K, \boldsymbol{\Sigma}_1, \ldots, \boldsymbol{\Sigma}_K) = \sum_{k=1}^{K} \pi_k \frac{L^{Ld}|\mathbf{Z}|^{L-d}}{\Gamma_d(L)|\boldsymbol{\Sigma}_k|^L} \exp\left\{-L \cdot \mathrm{tr}\left(\boldsymbol{\Sigma}_k^{-1}\mathbf{Z}\right)\right\} \tag{18}$$

where K is the number of subclasses in a class, π_k, $\boldsymbol{\Sigma}_k$ are the weight coefficient and cluster center of the kth subclass. The weight coefficient $\pi_1, \ldots, \pi_K$ are nonnegative and satisfies

$$\sum_{k=1}^{K} \pi_k = 1 \text{ and } \pi_k \geq 0. \tag{19}$$

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The EM algorithm is a widely used method for parameter estimation of mixture models [37], and we use it to estimate the parameters of WMM in the classification framework. Assuming $\mathbf{X} = \{\mathbf{Z}_1, \mathbf{Z}_2, \ldots, \mathbf{Z}_N\}$ are N training samples of a certain class, the specific steps of parameter estimation are as follows.

Step 1: Initialize parameters. K elements $\boldsymbol{\Sigma}_1^{(0)}, \ldots, \boldsymbol{\Sigma}_K^{(0)}$ are randomly selected as cluster centers, and the superscript (0) represents the parameter value at the $0th$ iteration. The initial weight coefficients of each subclass are equal, that is, $\pi_1^{(0)}, \ldots, \pi_K^{(0)} = 1/K$.

Step 2: For each training sample $\mathbf{Z}_n$, calculate its responsivity $\gamma_{kn}^{(i)}$ to the kth subclass according to the following:

$$\gamma_{kn}^{(i)} = \frac{\pi_k^{(i-1)} \mathcal{W}\left(\mathbf{Z}_n | \boldsymbol{\Sigma}_k^{(i-1)}\right)}{\sum_{k=1}^{K} \pi_k^{(i-1)} \mathcal{W}\left(\mathbf{Z}_n | \boldsymbol{\Sigma}_k^{(i-1)}\right)} \tag{20}$$

where $\mathcal{W}(\cdot)$ is the Wishart distribution function, and its PDF is (11).

Step 3: Update the cluster center $\boldsymbol{\Sigma}_k$ and weight coefficient π_k according to the following:

$$\boldsymbol{\Sigma}_k^{(i)} = \frac{\sum_{n=1}^{N} \gamma_{kn}^{(i)} \mathbf{Z}_n}{\sum_{n=1}^{N} \gamma_{kn}^{(i)}} \tag{21}$$

$$\pi_k^{(i)} = \frac{1}{N} \sum_{n=1}^{N} \gamma_{kn}^{(i)}. \tag{22}$$

Step 4: Repeat Steps 2 and 3 until the iterations converge. The iteration termination conditions are as follows:

$$\frac{1}{2}\mathrm{tr}\left(\boldsymbol{\Sigma}_k^{(i)} \cdot \mathrm{inv}\left(\boldsymbol{\Sigma}_k^{(i-1)}\right) + \boldsymbol{\Sigma}_k^{(i-1)} \cdot \mathrm{inv}\left(\boldsymbol{\Sigma}_k^{(i)}\right)\right) - d < \delta_{\boldsymbol{\Sigma}} \tag{23}$$

$$\left|\pi_k^{(i)} - \pi_k^{(i-1)}\right| < \delta_\pi \tag{24}$$

where $\mathrm{inv}(\cdot)$ represents matrix inversion, $\delta_{\boldsymbol{\Sigma}}$, and δ_π are the thresholds for cluster centers and weight coefficients. In this article, $K = 10$, $\delta_{\boldsymbol{\Sigma}}$, and δ_π are set to 10^{-3}. More details on parameter estimation are in [36].

The EM algorithm estimates parameters through an iterative process. Like most iterative methods, the EM algorithm needs to set initial values for parameters. However, different initial values often lead to different results. This means that the EM algorithm is only guaranteed to converge to a local optimal solution, not the global one [38], [39]. To mitigate the influence of the initial values on the final results, we set a number of different initial values for the parameters. After the iterative convergence of the EM algorithm, multiple final parameter values are obtained. Next, the data log-likelihood function is utilized to verify the goodness of fit of different parameters to the PolSAR data [40]. For the parameter $\Theta_m = \{\pi_{m1}, \ldots, \pi_{mK}, \boldsymbol{\Sigma}_{m1}, \ldots, \boldsymbol{\Sigma}_{mK}\}$, the PDF of covariance matrix $\mathbf{Z}$ can be calculated by

$$\begin{aligned} p(\mathbf{Z}|\Theta_m) &= \sum_{k=1}^{K} \pi_{\mathrm{mk}} p(\mathbf{Z}|\boldsymbol{\Sigma}_{\mathrm{mk}}) \\ &= \frac{L^{Ld}|\mathbf{Z}|^{L-d}}{\Gamma_d(L)} \sum_{k=1}^{K} \frac{\pi_{\mathrm{mk}} \exp\left\{-L \cdot \mathrm{tr}\left(\boldsymbol{\Sigma}_{\mathrm{mk}}^{-1}\mathbf{Z}\right)\right\}}{|\boldsymbol{\Sigma}_{\mathrm{mk}}|^L}. \end{aligned} \tag{25}$$

For the training sample set $\mathbf{X}$, the data log-likelihood function can be calculated by

$$\begin{aligned} L(\mathbf{X}|\Theta_m) &= \ln\left(\prod_{i=1}^{N} p(\mathbf{Z}_i|\Theta_m)\right) \\ &= NLd\ln L - N\ln\Gamma_d(L) + (L-d)\sum_{i=1}^{N}\ln|\mathbf{Z}_i| \\ &\quad + \sum_{i=1}^{N}\ln\left(\sum_{k=1}^{K}\pi_{\mathrm{mk}}\exp(-L\cdot d_W(\mathbf{Z}_i, \boldsymbol{\Sigma}_{\mathrm{mk}}))\right) \end{aligned} \tag{26}$$

where $d_W(\mathbf{Z}_i, \boldsymbol{\Sigma}_{\mathrm{mk}})$ is the Wishart distance (14). The first three terms of (26) do not change with the parameters. Therefore, (26) can be simplified as

$$L(\mathbf{X}|\Theta_m) = \sum_{i=1}^{N}\ln\left(\sum_{k=1}^{K}\pi_{\mathrm{mk}}\exp(-L\cdot d_W(\mathbf{Z}_i, \boldsymbol{\Sigma}_{\mathrm{mk}}))\right). \tag{27}$$

Finally, the parameters of WMM are determined as the arguments of the maxima of the data log-likelihood function and will be used to calculate the distance measure in the unified classification framework.

B. Classification Based on Distance Measures

In the unified framework, each sample is classified into the class with the smallest distance measures. The Wishart, Bartlett, RW, and SRW distance measures characterize the similarity between data obeying the complex Wishart distribution. However, in the unified framework, we assume that PolSAR data obey the WMM distribution rather than the complex Wishart distribution. Therefore, these four distance measures cannot be used directly for classification. WMM considers that there are multiple subclasses in a single class, and each subclass still obeys the Wishart distribution. Therefore, we can use (14)–(17) to get the Wishart, Bartlett, RW, and SRW distance measures from the sample to each subclass in the class. In the framework, we refer to these four distance measures as the sample-to-subclass distances. The classification framework assigns samples to different classes, not subclasses. Therefore, what we need to obtain is the distance measure from the sample to each class, which we call the sample-to-class distance in this framework. According to the relationship between WMM and the Wishart model, we propose two schemes to obtain the sample-to-class distance.

Just as we often define the distance from a point x to a set of points G as the minimum value from x to the elements in G, the first scheme is to take the minimum value of the sample-to-subclass distances as the sample-to-class distance. It should be noted that the sample-to-subclass distance needs consider the influence of the weight coefficient, which represents the prior probability that the sample belongs to the subclass. Therefore, the first scheme can be defined as

$$d_{\mathrm{WMM1}}(Z, \Theta) = \min_{k=1,\ldots,K}(d(\mathbf{Z}, \boldsymbol{\Sigma}_k) + \ln(\pi_k)). \tag{28}$$

Since the distance measure is usually derived by taking the negative logarithm of the likelihood function, the sample-to-subclass distance can be transformed into the likelihood

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probability. For WMM, the likelihood probability that the sample belongs to the class is obtained by weighted summation of the likelihood probabilities that the sample belongs to all subclasses in the class. After taking the negative logarithm of the obtained likelihood probability, we can get the sample-to-class distance. Therefore, the second scheme is defined as

$$d_{\mathrm{WMM2}}(Z,\Theta) = -\ln\left(\sum_{k=1}^{K}\pi_k \cdot e^{-d(\mathbf{Z},\boldsymbol{\Sigma}_k)}\right). \tag{29}$$

In (28) and (29), $\Theta = \{\pi_1,\ldots,\pi_K,\boldsymbol{\Sigma}_1,\ldots,\boldsymbol{\Sigma}_K\}$ are the parameters of WMM, estimated by the EM algorithm and the data log-likelihood function. $d(\mathbf{Z},\boldsymbol{\Sigma}_k)$ is the sample-to-subclass distance measure. In the unified classification framework, we take the Wishart, Bartlett, RW, and SRW distance measures as $d(\mathbf{Z},\boldsymbol{\Sigma}_k)$. Among the combinations of the four sample-to-subclass distance measures and the two schemes, the one with the best classification performance is selected as the final result for the unified framework.

For multifrequency data, the cross-correlation between different frequency bands is much smaller than the correlation of each frequency band itself [29]. Therefore, the data between different frequency bands are generally assumed to be statistically independent. The distance measure for multifrequency PolSAR data can be written as

$$d_J(Z_1,\ldots,Z_J,\Theta_1,\ldots,\Theta_J) = \sum_{j=1}^{J} d\left(\mathbf{Z}_j,\Theta_j\right) \tag{30}$$

where J represents the number of frequency bands, and Z_j and Θ_j are the j band covariance matrix and the corresponding model parameters, respectively. $d(\mathbf{Z}_j,\Theta_j)$ represents the distance measure in the j band. However, some researchers consider that the cross-correlation information between different frequency bands plays an important role in classification tasks [26], [41]. Ferro-Famil et al. [26] hold the view that it is necessary to simultaneously consider the cross-correlation information between different frequency bands. To make effective use of PolSAR data between frequency bands, we do not assume that the data are statistically independent in the proposed unified classification framework. Since the $\mathbf{C}$ matrix (8) contains all the information of two frequency bands, the distance measures for single-frequency data are applied to obtain the dual-frequency distance measure in the unified framework.

IV. Experiments and Results

A. Classification Capabilities of FP, CP, and DP Modes

To enable the unified classification framework to effectively classify PolSAR data, we first evaluate the performance of two schemes (28) and (29) under different sample-to-subclass distance measures. The image of San Francisco area acquired by the C-band GF-3 satellite in the FP strip mode is used for the experiment. It was obtained at 01:56 on September 15, 2017, at an incident angle of 21°. There are a total of five classes in the scene, including water, vegetation, and three types of urban areas. The governing factor defining the three urban areas is the rotation angle. All three urban classes contain rotated buildings, with a uniform rotation angle within a class but different rotation angles relative to each other [42]. The image size is 1600 × 2500, and the corresponding Pauli color-coded image, optical remote-sensing image, and ground-truth map are shown in Fig. 2(a)–(c). The ground-truth information is collected by Google Maps with high resolution in the same period. After 5 × 5 multilook processing, the Wishart, Bartlett, RW, and SRW distance measures are used for classification. The overall accuracy (OA) indicator is utilized to evaluate the classification performance, and the specific results are given in Table I. In the table, WMM-1 and WMM-2 represent schemes (28) and (29), respectively. To verify that WMM has a better performance than the Wishart model, we also give the results of the scheme assuming the data obey the Wishart distribution, denoted by "–" in Table I.

TABLE I
OA (%) of Two Schemes Under Four Distance Measures on San Francisco Area

Scheme	Distance Measures			
	Wishart	Bartlett	RW	SRW
-	87.92	84.70	79.24	84.54
WMM-1	92.41	92.16	91.96	92.66
WMM-2	92.85	91.60	91.24	92.73

Table I shows that no matter which distance measure is used, the OAs of both WMM schemes are higher than the scheme assuming the data obey the Wishart distribution. This verifies that WMM can more accurately characterize the statistical distribution than the Wishart model. The classification performance of two WMM schemes is close under the four distance measures. Scheme (28) performs slightly better under the Bartlett and RW distance measures, and scheme (29) performs slightly better under the Wishart and SRW distance measures. Since scheme (29) has the highest OA of 92.85% under the Wishart distance measure in the whole experiment, we choose this combination as default in the proposed classification framework.

Next, we will compare the classification capabilities of FP, CP, and DP modes under the unified framework. Here, the CP mode includes HCP, $\pi/4$. The DP mode includes HH-HV and VV-HV. Both DCP and HCP transmit circularly polarized signals and receive echoes in two orthogonally polarized directions. The only difference between DCP and HCP is the received basis. The data of DCP and HCP can be transformed equivalently [43], and the classification results are also the same. Therefore, the classification performance of DCP is not given here. HH-VV mode does not belong to the single-transmit and double-receive system and is not discussed here. Four indicators, including the user's accuracy (UA), the producer's accuracy (PA) [44], OA, and Kappa coefficient, are utilized to analyze the classification capabilities. Fig. 2(d)–(h) show classification images of the San Francisco area. The detailed classification results are given in Table II.

The classification results demonstrate that the FP mode has the highest OA and Kappa coefficient, which are 92.85% and 0.8779, respectively. For all five classes, the UAs of

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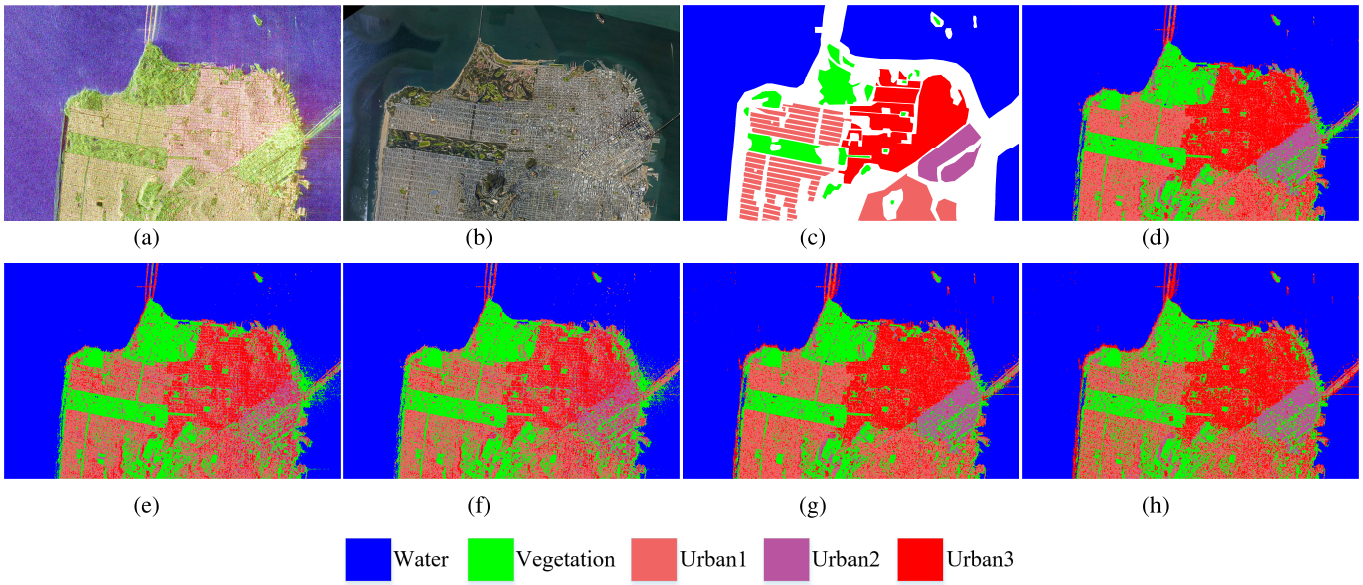


Fig. 2. (a)–(c) Experimental data of San Francisco area. (d)–(h) Corresponding classification images under the unified classification framework. (a) Pauli RGB. (b) Optical remote-sensing image. (c) Ground reference map. (d) FP. (e) HCP (CP). (f) $\pi/4$ (CP). (g) HH-VH (DP). (h) VV-HV (DP).

TABLE II
CLASSIFICATION ACCURACY (%) OF DIFFERENT POLARIZATION MODES ON THE SAN FRANCISCO AREA UNDER THE UNIFIED FRAMEWORK

Class		FP	CP		DP	
			HCP	$\pi/4$	HH-VH	VV-HV
Water	UA	99.99	99.98	99.99	99.93	99.97
	PA	99.98	99.88	99.82	99.84	99.88
Vegetation	UA	80.27	70.19	68.64	75.65	63.00
	PA	91.78	90.14	89.23	90.21	84.15
Urban1	UA	79.36	61.59	63.07	74.67	67.71
	PA	80.20	61.09	64.08	75.17	61.83
Urban2	UA	79.87	28.39	34.16	75.12	64.12
	PA	80.20	47.55	58.17	77.97	72.89
Urban3	UA	85.95	74.34	75.68	82.44	80.20
	PA	79.18	52.00	50.26	73.92	71.61
OA(%)		92.85	84.79	85.26	91.10	88.30
Kappa		0.8779	0.7418	0.7500	0.8481	0.8007

the FP mode are the highest among all polarization modes, so are the PAs. This demonstrates that the FP mode has better performance than CP and DP modes in classification. Fig. 2(d) also shows that the FP mode can distinguish the five classes well. The OA of HH-VH mode is 91.10% which is only 1.75% lower than the FP mode. The VV-HV mode has an OA of 88.30%, which is 4.55% lower than the FP mode. Overall, the classification capability of the DP mode is close to the FP mode, and the OA is about 2%–5% lower than the FP mode. The HCP and $\pi/4$ modes have the OA of 84.79% and 85.26%, which are 8.06% and 7.59% lower than the FP mode. Compared with the DP mode, the classification capability of the CP mode is worse.

From Table II, it can be seen that the UAs, PAs of HCP and $\pi/4$ modes are almost all lower than the corresponding UAs, PAs of FP, HH-VH, and VV-HV modes for three urban classes. Especially for the urban 2 class, the UAs of HCP and $\pi/4$ modes are only 28.39% and 34.16%, which indicates that a large number of samples from other classes are misclassified as urban 2. The corresponding PAs are only 47.55% and 58.17%. Fig. 2(e) and 2(f) shows that three types of urban areas are erroneously mixed. For urban 1 and urban 3 classes, misclassified pixels are mainly labeled as other urban classes. For the urban 2 class, some of the misclassified pixels are labeled as other urban classes, and some are labeled as vegetation class. The reason is that the rotation angle of the urban 2 class is large, and there is a certain volume scattering compared to the other two urban classes. Compared with two CP modes, two DP modes perform well in distinguishing the urban region. Take HCP and HH-VH modes as examples. The UAs of the HH-VH mode are 74.67%, 75.12%, and 82.44% for urban 1, urban 2, and urban 3 classes, respectively, which are 13.08%, 46.73%, and 8.1%, respectively, higher than the corresponding UAs of the HCP mode. The PAs of the HH-VH mode are 75.17%, 77.97%, and 73.92% for urban 1, urban 2, and urban 3 classes, respectively, which are 14.08%, 30.42%, and 21.92% higher than the corresponding PAs of the HCP mode, respectively. It can be seen that the HCP model cannot distinguish the three urban classes well. A possible explanation is that there are a large number of dihedral structures in urban areas, and the HCP mode is not sensitive to dihedral structures with different rotation angles. This is consistent with the analysis in [45] that the HCP model has difficulty classifying buildings with different orientations. Overall, the classification performance of the HH-VH model in the urban region is better than the HCP model, which is the main reason that the OA of the HH-VH model is 6.31% higher than that of the HCP model.

Next, another PolSAR data is utilized to analyze the classification capabilities of FP, CP, and DP modes. The data is an L-band image of Boao town, Hainan province, China.

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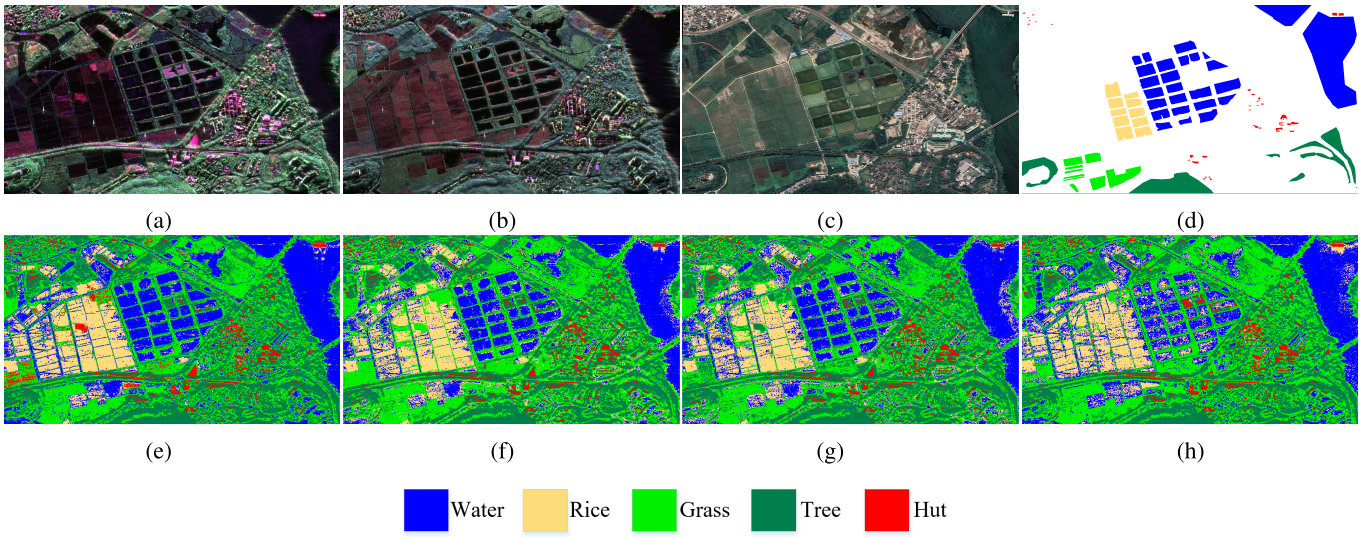


Fig. 3. (a)–(d) Experimental data of Boao town. (e)–(h) Corresponding classification images under the unified classification framework. (a) Pauli RGB (L-band). (b) Pauli RGB (X-band). (c) Optical remote-sensing image. (d) Ground reference map. (e) FP. (f) HCP (CP). (g) $\pi/4$ (CP). (h) HH-VH (DP).

TABLE III
CLASSIFICATION ACCURACY (%) OF DIFFERENT POLARIZATION MODES ON THE L-BAND BOAO DATA UNDER THE UNIFIED FRAMEWORK

Class		FP	CP		DP	
			HCP	$\pi/4$	HH-VH	VV-HV
Water	UA	99.15	98.24	97.97	97.26	95.21
	PA	95.51	91.95	91.32	82.30	78.03
Rice	UA	82.88	70.33	67.72	49.83	41.61
	PA	96.14	91.11	90.42	87.98	79.26
Grass	UA	69.12	65.69	67.61	65.29	66.98
	PA	91.64	88.78	88.96	89.55	90.47
Tree	UA	96.45	94.88	95.12	95.15	95.50
	PA	85.36	83.20	83.67	82.84	82.87
Hut	UA	97.78	93.59	92.60	96.03	77.54
	PA	99.32	97.90	97.63	98.47	92.47
OA(%)		93.30	89.94	89.60	83.86	80.34
Kappa		0.8888	0.8352	0.8298	0.7463	0.6947

It was collected in the FP strip mode at 16:41 on February 25, 2021, at an incident angle of 60°, with a size of 1400 × 2500. The corresponding Pauli color-coded image, optical remote-sensing image, and ground-truth map are shown in Fig. 3(a), (c), and (d). The ground-truth information is collected through field surveys. The image contains five classes, including water, rice, grass, tree, and hut. The water class contains ponds and rivers, and the hut class is scattered along the road. The 5 × 5 multilook data is used as the input of the unified classification framework. Fig. 3(e)–(h) shows the corresponding classification images. The detailed classification results of the Boao town are given in Table III.

From Table III, it can be seen that the FP mode still has the highest OA and Kappa coefficient, which are 93.30% and 0.8888, respectively. The UAs of the FP mode are 99.15%, 82.88%, 69.12%, 96.45%, and 97.78% for water, rice, grass, tree, and hut, respectively. The corresponding PAs are 95.51%, 96.14%, 91.64%, 85.36%, and 99.32%, respectively, which are the highest among all polarization modes. The OAs of HCP and $\pi/4$ modes are 89.94% and 89.60%, respectively, which are 3.36% and 3.7% lower than the FP mode, respectively. The HH-VH and VV-HV modes have the worst performance with the OAs of 83.86% and 80.34%, respectively, which are 9.44% and 12.96% lower than the FP mode, respectively.

Fig. 3 shows that FP, CP, and DP modes have difficulties in distinguishing water and rice classes. The main reason is that the rice has just been planted in paddy fields and is in the early stages of growth. Electromagnetic waves in the L-band can easily penetrate the leaves of rice and reach the water surface [46]. Therefore, rice and water classes have similar scattering information in the L-band. For the rice class, the UA of the FP mode is the highest among the five polarization modes. The UAs of HCP and $\pi/4$ modes are 70.33% and 67.72%, respectively, which are 12.55% and 15.16% lower than the FP mode, respectively. The UAs of HH-VH and VV-HV modes are 49.83% and 41.61%, respectively, which are 33.05% and 41.27% lower than the FP mode, respectively. Likewise, the PA of the FP mode is the highest among the five polarization modes for the rice class. The PAs of HCP and $\pi/4$ modes are 91.11% and 90.42%, respectively, which are 5.03% and 5.72% lower than the FP mode, respectively. The PAs of HH-VH and VV-HV modes are 87.98% and 79.26%, respectively, which are 8.16% and 16.88% lower than the FP mode, respectively. For the rice class, the UAs and PAs of CP and DP modes are reduced compared to the FP mode, while the CP mode does not reduce as much as the DP mode. A similar result can also be drawn from the classification results of the water class. One possible reason is that in the L-band, the DP mode is more sensitive to water surface than the CP mode. The water surface is mostly specular reflection, and the backscatter coefficient is extremely low. When the water surface is disturbed, the backscatter received in the DP mode is stronger and more likely to be

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TABLE IV

OA (%) OF SINGLE-FREQUENCY AND DUAL-FREQUENCY SAR ON THE BOAO TOWN UNDER THE UNIFIED FRAMEWORK

Band	FP	CP		DP	
		HCP	$\pi/4$	HH-VH	VV-HV
L	93.30	89.94	89.60	83.86	80.34
X	91.74	88.02	87.85	89.07	88.80
LX-1	96.14	94.32	94.36	94.22	94.77
LX-2	96.85	95.85	95.79	95.82	95.99

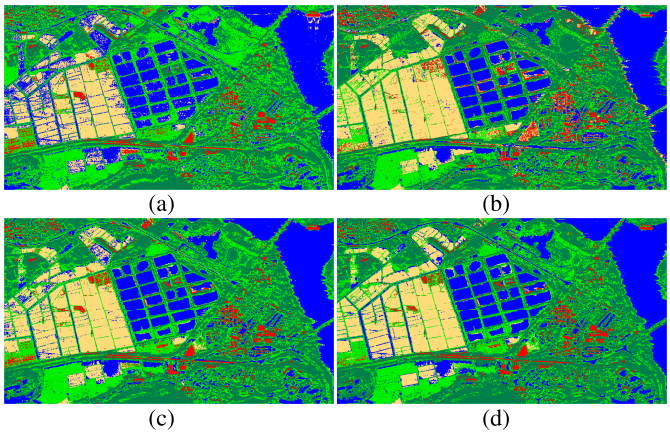


Fig. 4. Classification images of single-frequency and dual-frequency SAR in the FP mode on the Boao town under the unified framework. (a) L. (b) X. (c) LX-1. (d) LX-2.

confused with the backscatter of rice. In Fig. 3, it can be seen that more water pixels are misclassified as rice in the DP mode.

From the experimental results of the above two datasets, it can be concluded that the FP mode has the highest classification accuracy and Kappa coefficient in various scenarios. This is consistent with the theoretical analysis. FP data contain the complete scattering information of the target, while CP and DP data are a subset of FP data. At present, the CP model has attracted much attention. Some researchers believe that the CP mode is a good alternative when the FP mode is not possible. However, the working way of CP and DP modes are both single-transmit and double-receive, respectively. Although the information contained in the CP and DP data is different, both can be represented by a second-order covariance matrix. The experimental results show that the CP mode may have better classification performance than DP in certain scenarios, and in some scenarios, the situation may be reversed.

B. Classification Capabilities of Dual-Frequency SAR

In this section, we will compare the classification capabilities of dual-frequency SAR in FP, CP, and DP modes under the unified framework. To fully exploit the classification capabilities of dual-frequency SAR, we also evaluate the impact of the cross-correlation information between different frequency bands on the classification. First, the L- and X-band data of Boao town are utilized as experimental data. The L- and X-band data were acquired simultaneously by two onboard SAR systems. The noise equivalent sigma zero (NESZ) values of both SAR systems are below -40 dB. Before the experiment, the two-band data were filtered with a 5×5 boxcar filter. Table IV gives the OA of single-frequency and dual-frequency SAR in five polarization modes under the proposed framework. The L and X in the first column of the table represent the classification results of single-frequency SAR. The LX-1 and LX-2 represent the classification results of dual-frequency SAR. The LX-1 indicates that in classification tasks, the dual-frequency distance measure is obtained by directly summing the two single-frequency distance measures, that is, ignoring the cross-correlation information. The LX-2 indicates that in classification tasks, the dual-frequency distance measure is obtained by constructing a multidimensional covariance matrix. In this way, the cross-correlation information between different frequency bands can be fully exploited. The classification results of the L-band data have been given in Section IV-A. However, to directly compare the classification accuracy between different frequency bands, the corresponding results are still given here.

TABLE V

CLASSIFICATION ACCURACY (%) OF SINGLE-FREQUENCY AND DUAL-FREQUENCY SAR IN THE FP MODE ON THE BOAO TOWN UNDER THE UNIFIED FRAMEWORK

Class		Band			
		L	X	LX-1	LX-2
Water	UA	99.15	99.24	99.80	99.95
	PA	95.51	95.69	98.16	98.32
Rice	UA	82.88	91.03	94.71	94.71
	PA	96.14	97.57	99.19	99.78
Grass	UA	69.12	61.74	74.43	80.07
	PA	91.64	90.20	94.31	93.03
Tree	UA	96.45	89.45	97.40	96.84
	PA	85.36	77.55	88.91	92.17
Hut	UA	97.78	75.16	98.20	98.41
	PA	99.32	87.79	99.23	98.97
OA(%)		93.30	91.74	96.14	96.85
Kappa		0.8888	0.8630	0.9353	0.9470

For the five polarization modes in Table IV, the OAs of LX-1 and LX-2 are higher than the OAs of L and X. This illustrates that classification capability can be improved by combining two different single-frequency data. The OA of LX-2 is approximately 1%–2% higher than that of LX-1, which illustrates that the cross-correlation information between different frequency bands is helpful for classification. Compared to the higher OA in L and X, the OA of LX-2 is improved by 3.55%, 5.91%, 6.19%, 6.75%, and 7.19% in FP, HCP, $\pi/4$, HH-VH, and VV-HV modes, respectively. Next, we take the FP mode as an example for further analysis. Fig. 4(a)–(d) shows the classification images of single-frequency and dual-frequency SAR in the FP mode, and Table V gives the detailed classification results.

From Table V, it can be seen that the OAs of L- and X-band are 93.30% and 91.74%, respectively. For the water and rice classes, the UA and PA of the X-band are higher than those of the L-band. Compared with the L-band, the

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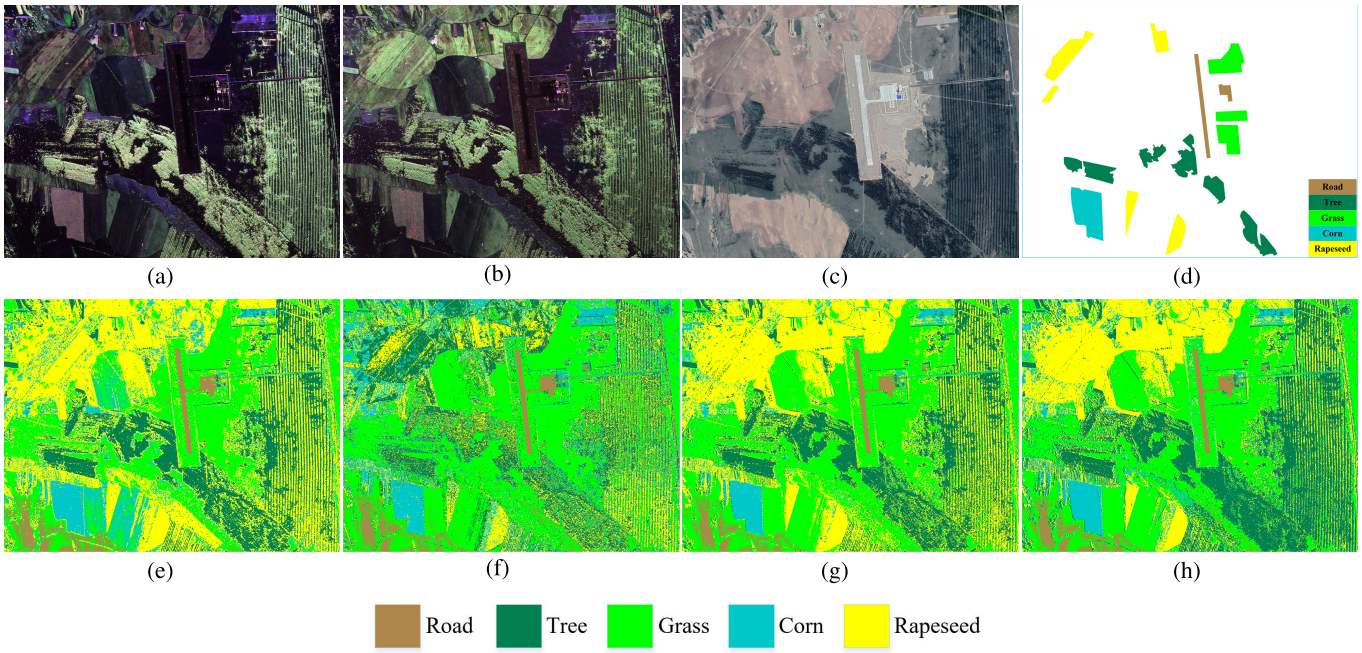


Fig. 5. (a)–(d) Experimental data of Yudaokou town. (e)–(h) Corresponding classification images under the unified classification framework. (a) Pauli RGB (L-band). (b) Pauli RGB (S-band). (c) Optical remote-sensing image. (d) Ground reference map. (e) L. (f) S. (g) LS-1. (h) LS-2.

electromagnetic waves in the X-band can hardly penetrate the rice leaves to reach the water surface. Therefore, water and rice classes can be easily distinguished in the X-band. For grass, tree, and hut classes, L-band has a better performance for its deeper penetration. In general, L- and X-band have their own advantages in classification. The OA of LX-1 is 96.14%, which is 0.71% lower than that of LX-2. This indicates that the cross-correlation information between two frequency bands is beneficial to improve the classification capability. The OA of LX-2 is 96.85%, which is 3.55% and 5.11% higher than L- and X-band, respectively. Specific to the classification results of each class, the UA and PA of LX-2 are always close to or higher than those of the L- and X-band. The results illustrate that dual-frequency SAR inherits and develops the advantages of two single-frequency SAR in classification.

Next, we will compare the classification capabilities of dual-frequency data with L- and S-band data. The L- and S-band data were acquired simultaneously by two onboard SAR systems in the area of Yudaokou town, Hebei province, China. They were collected in the FP strip mode at 15:19 on September 27, 2021, at 55° incident angle. The NESZ values of both SAR systems are below −40 dB. The image has a size of 1800 × 2400 and there are a total of five classes, including road, tree, corn, grass, and rapeseed. The corresponding Pauli color-coded images, optical remote-sensing image, and ground-truth map are shown in Fig. 5(a)–(d). The ground-truth information is collected by field surveys. Before the experiment, the two-band data were filtered with a 5 × 5 boxcar filter. Table VI gives the OA of single-frequency and dual-frequency SAR in five polarization modes under the proposed framework. In the table, L and S represent the classification results of single-frequency SAR. LS-1 and LS-2 represent the classification results of dual-frequency SAR. LS-1 indicates that the dual-frequency distance measure is obtained by directly summing the two single-frequency distance measures. LS-2 indicates that the dual-frequency distance measure is obtained by constructing a multidimensional covariance matrix.

TABLE VI
OA (%) OF SINGLE-FREQUENCY AND DUAL-FREQUENCY SAR ON THE YUDAOKOU TOWN UNDER THE UNIFIED FRAMEWORK

Band	FP	CP		DP	
		HCP	$\pi/4$	HH-VH	VV-HV
L	93.18	86.22	85.22	91.14	88.83
S	81.42	73.23	70.67	78.74	75.51
LS-1	96.16	91.90	90.69	95.10	95.05
LS-2	97.94	93.89	93.27	97.12	96.74

For all five polarization modes in Table VI, the OA of the L-band is more than 10% higher than that of the S-band. The reason is that most of the scene is covered by vegetation, and the L-band penetrates deeper into the vegetation than the S-band. This illustrates that frequency plays a very important role in classification tasks. For the five polarization modes, the classification accuracy further improved by combining the L- and S-band data. The OA of LS-2 is approximately 2%–2.5% higher than that of LS-1, which proves again that the cross-correlation information between two frequency bands is helpful for classification. Compared to the OA of L, the OA of LS-2 is improved by 4.76%, 7.67%, 8.05%, 5.98%, and 7.91% in FP, HCP, $\pi/4$, HH-VH, and VV-HV modes, respectively. Next, we take the FP mode as an example for further analysis. Fig. 5(e)–(h) shows the classification images of single-frequency and dual-frequency SAR in the FP mode, and Table VII gives the detailed classification results.

From Table VII, it can be seen that the OA of the L-band is 11.76% higher than that of the S-band. For the road class,

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TABLE VII
CLASSIFICATION ACCURACY (%) OF SINGLE-FREQUENCY AND DUAL-FREQUENCY SAR IN THE FP MODE UNDER THE UNIFIED FRAMEWORK

Class		Band			
		L	S	LS-1	LS-2
Road	UA	99.87	99.99	100	100
	PA	99.81	99.81	99.98	99.99
Tree	UA	97.77	77.57	96.88	98.93
	PA	92.22	70.57	92.43	96.03
Grass	UA	99.84	99.76	99.97	99.99
	PA	99.23	99.71	99.77	99.73
Corn	UA	81.80	81.90	97.55	96.70
	PA	96.03	95.46	98.16	98.17
Rapeseed	UA	88.91	66.13	90.63	95.41
	PA	85.99	66.87	95.74	98.30
OA(%)		93.18	81.42	96.16	97.94
Kappa		0.9112	0.7582	0.9498	0.9731

the classification performance of L- and S-band is close. The reason is that the road is flat and impenetrable. There is little difference between scattering information in L- and S-band. For tree and rapeseed classes, the L-band has better classification performance than the S-band for its deeper penetration. For the grass class, the classification performance is close because both L- and S-band can fully penetrate low vegetation. Since the corn is at maturity and has few leaves, the L- and S-band can easily penetrate to the bottom. Therefore, L- and S-band have similar classification performance for the corn class. From the above analysis, it can be concluded that observation frequency has a very important impact on the classification results. The OA of LS-1 is 96.16%, which is 1.78% lower than that of LS-2. This again indicates that cross-information is helpful to improve classification accuracy. The OA of LS-2 is 97.94%, which is 4.76% and 16.52% higher than those of the L- and S-band, respectively. For the five classes, the UAs and PAs of LS-2 are higher than those of the L- and S-band. This demonstrates that the classification capability is further improved by combining the L- and S-band data.

From the classification results of the above two datasets, it can be concluded that the classification capabilities of the three modes are all improved by combining two single-frequency data. By combining two single-frequency data, the OA is improved by about 4%–5% in the FP mode and 6%–8% in the CP/DP mode. The dual-frequency SAR inherits and develops the advantages of two single-frequency SAR in classification. For a scene of interest, single-frequency data may not be able to effectively characterize the scattering information of all classes. Therefore, the combination of data in different frequency bands is an effective method to improve classification capability.

V. CONCLUSION

This article proposed a unified framework to compare the classification capabilities of multipolarization and dual-frequency SAR. To uniformly process multipolarization and dual-frequency SAR data, the unified framework is directly based on the covariance matrix for classification tasks without using polarimetric indicators. Additionally, WMM is utilized to accurately describe the statistical distribution of PolSAR data. The parameters of WMM are estimated by the EM algorithm in the unified framework. To mitigate the influence of the initial values on the final parameters, the data log-likelihood function is utilized to verify the goodness of fit of different parameters to PolSAR data and determine the final parameters of WMM. Then, among the combinations of four sample-to-subclass distances and two schemes for obtaining sample-to-class distances, the one with the best classification performance is selected as the default for this framework.

The classification capabilities of FP, CP, and DP modes are first compared through the unified framework. Experimental results demonstrate that the FP mode has the highest classification accuracy in various scenarios. The classification capabilities of CP and DP modes are close to the FP mode in some specific classes. For CP and DP modes, neither one always has better classification performance than the other in various scenarios. The results show that the CP mode is superior to the DP model in distinguishing water and rice classes, and the OA is about 4% lower than the FP mode. The DP mode has better classification results in distinguishing urban regions, and the OA is about 2%–5% lower than the FP mode. Then, the classification capability of dual-frequency SAR is compared through the proposed framework. Experimental results show that the cross-correlation information between different frequency bands helps to improve the OA by about 1%–2.5%. Compared with single-frequency SAR, the OA of dual-frequency SAR is improved by 4%–5% in the FP mode, and 6%–8% in CP and DP modes. By combining different single-frequency data, the advantages of each frequency band in distinguishing ground objects are inherited and developed. The comparison results can provide a reference for the optimal selection of polarization modes and frequency bands of the existing multipolarization and dual-frequency SAR payloads and the design of future SAR systems. In the future, more multipolarization and dual-frequency SAR data are needed to draw general conclusions about the classification capability.

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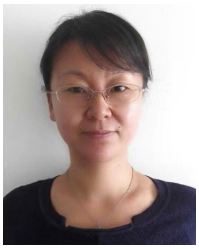
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